

MSI cw: helical vs axial CT

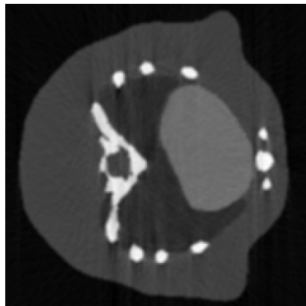
Piotr Blaszyk

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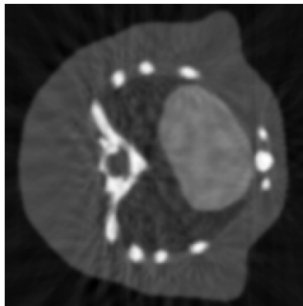
January 29, 2025

Results

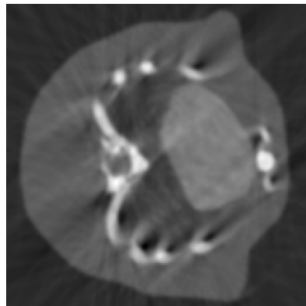
axial



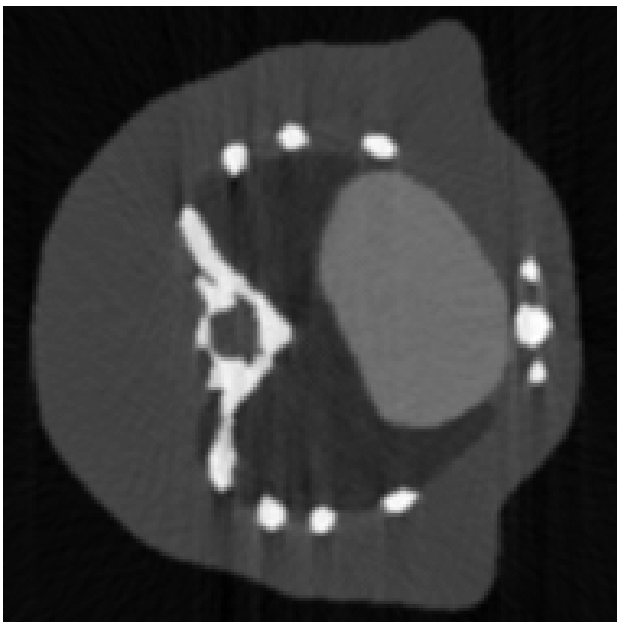
helical 1.0



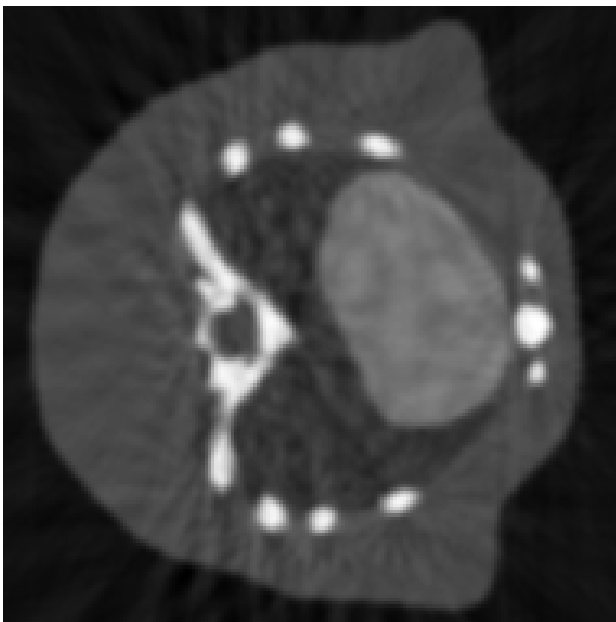
helical 2.0



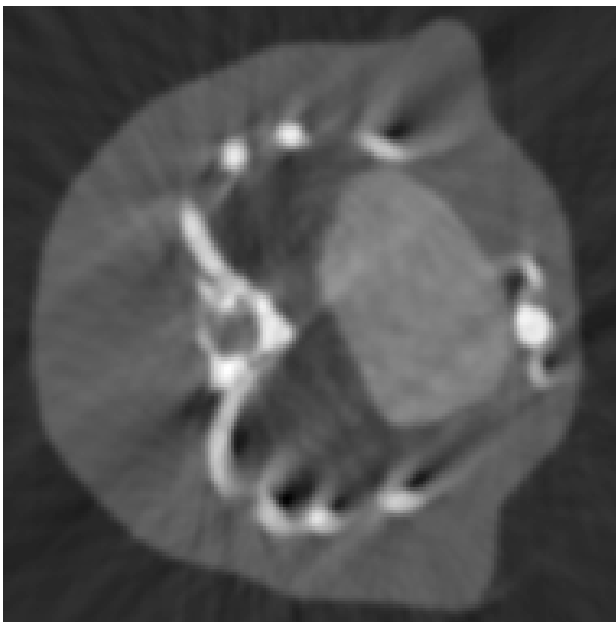
Axial



Helical 1.0



Helical 2.0



Assumptions

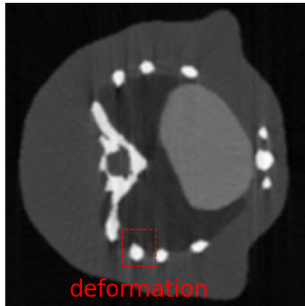
- ▶ half-scan
- ▶ 1 projection $\sim 1^\circ$
- ▶ no gaps between axial scans

Analysis: qualitative

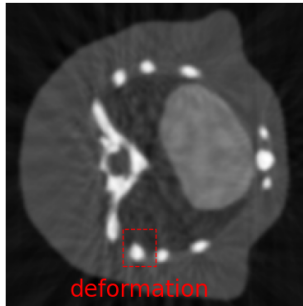
	axial	helical 1.0	helical 2.0
streak	low	medium	high
blurring	low	medium	high
deformations	low	medium	high

Analysis: qualitative

axial



helical 1.0



helical 2.0



Analysis: quantitative

	axial	helical 1.0	helical 2.0
SNR	35	14	16
CNR	46	21	13
RMSE	0%	4%	12%

ROIs

axial

helical 1.0

helical 2.0

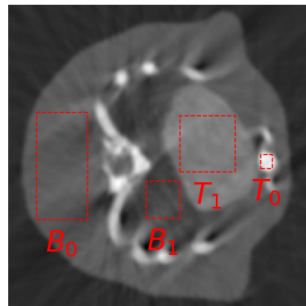
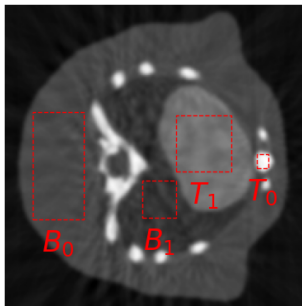
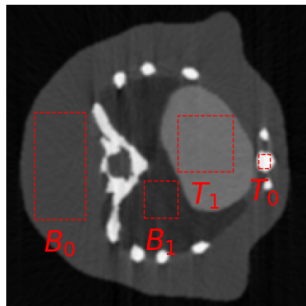


Figure: Background: B_0 , B_1 ; target: T_0 , T_1 .

Helical CT: pros

- ▶ Faster
- ▶ Higher z-axis resolution
- ▶ Reduced motion artifacts
- ▶ Increased scanner lifespan
- ▶ Reduced requirements for contrast material

Helical CT: cons

- ▶ Interpolation artifacts
- ▶ Lower x- and y-axis resolution

Helical CT: applications

- ▶ Trauma
- ▶ Pediatric
- ▶ Vascular (pulmonary embolism, aneurysms, stroke)
- ▶ Lung cancer
- ▶ Orthopedic and spine
- ▶ Contrast (abdominal, pelvic, cardiac)
- ▶ Head (brain, skull base, inner ear, sinus)

Questions?